```
#include <stdio.h>
int main() {
    int n, m, i, j, k;
    printf("1WA24CS217\n");
    printf("Enter number of processes: ");
    scanf("%d", &n);
    printf("Enter number of resources: ");
    scanf("%d", &m);
    int allocation[n][m];
    int max[n][m];
    int need[n][m];
    int available[m];
    int work[m];
    int finish[n];
    int safeSequence[n];
    printf("\nEnter Allocation Matrix:\n");
    for(i = 0; i < n; i++) {
        for(j = 0; j < m; j++) {
            scanf("%d", &allocation[i][j]);
        }
    }
    printf("\nEnter Maximum Demand Matrix:\n");
    for(i = 0; i < n; i++) {
        for(j = 0; j < m; j++) {
            scanf("%d", &max[i][j]);
        }

    printf("\nEnter Available Resources:\n");
    for(i = 0; i < m; i++) {
        scanf("%d", &available[i]);
```

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    work[i] = available[i];
}
for(i = 0; i < n; i++) {
    for(j = 0; j < m; j++) {
        need[i][j] = max[i][j] - allocation[i][j];
    }
}
for(i = 0; i < n; i++) {
    finish[i] = 0;
}
int count = 0;
while(count < n) {
    int found = 0;
    for(i = 0; i < n; i++) {
        if(finish[i] == 0) {
            int possible = 1;
            for(j = 0; j < m; j++) {
                if(need[i][j] > work[j]) {
                    possible = 0;
                    break;
                }
            }
            if(possible)
                for(k = 0; k < m; k++) {
                    work[k] += allocation[i][k];
                }
                safeSequence[count] = i;
                count++;
                finish[i] = 1;
                found = 1;
```

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            }
        }
    }

    if(found == 0)
        break;
}
int safe = 1
for(i = 0; i < n; i++) {
    if(finish[i] == 0) {
        safe = 0;
        break;
    }
}
if(safe) {
    printf("\nSystem is in a safe state.\n");
    printf("Safe sequence is: ");

    for(i = 0; i < n; i++) {
        printf("P%d", safeSequence[i]);

        if(i != n - 1)
            printf(" -> ");
    }

    printf("\n");
}
else {
    printf("\nSystem is not in a safe state.\n");
}
```

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    return 0;
}
```

OUTPUT:

```
1WA24CS217
Enter number of processes: 5
Enter number of resources: 3

Enter Allocation Matrix:
0 1 0
2 0 0
3 0 2
2 1 1
0 0 2

Enter Maximum Demand Matrix:
7 5 3
3 2 2
9 0 2
2 2 2
4 3 3

Enter Available Resources:
3 3 2

System is in a safe state.
Safe sequence is: P1 -> P3 -> P4 -> P0 -> P2

Process returned 0 (0x0)   execution time : 66.224 s
```